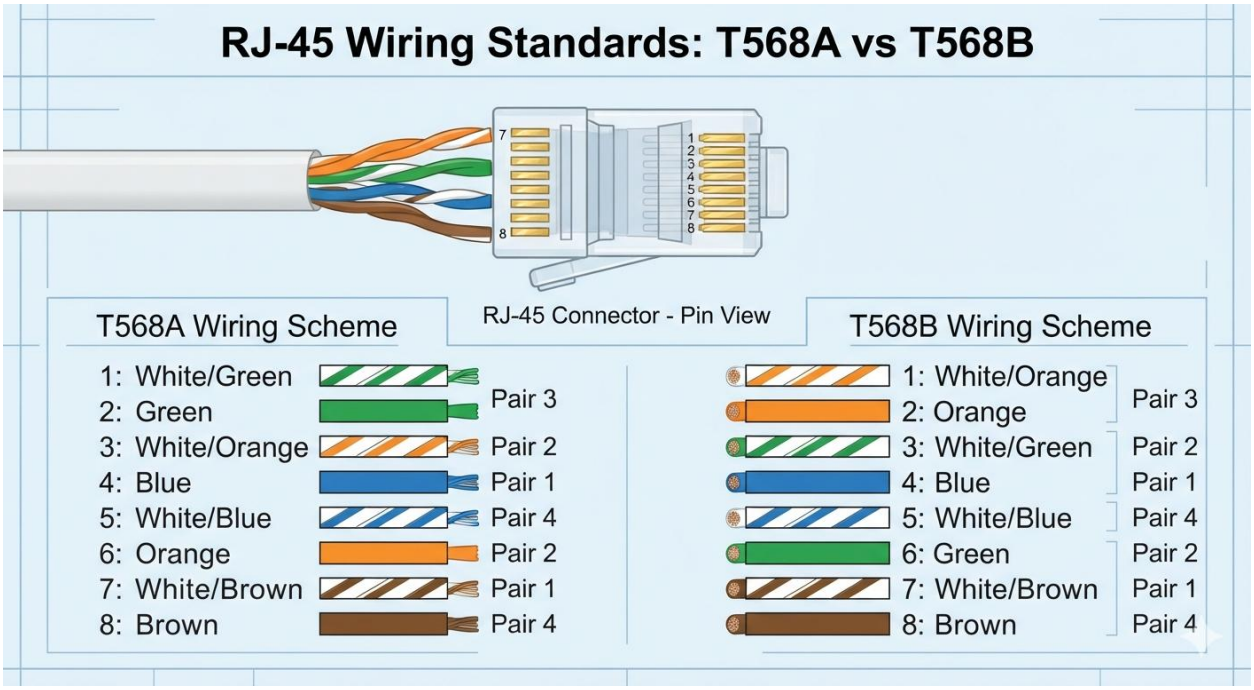
1. **Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.**



2. **Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.**

Ethernet Standard	Speed	Cable Type	Maximum Cable Length	Connector Type
10BASE-T	10 Mbps	Cat3 or better UTP	100 m	RJ-45
100BASE-TX	100 Mbps	Cat5 or better UTP	100 m	RJ-45
1000BASE-T	1 Gbps	Cat5e or better UTP	100 m	RJ-45

3. **Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.**

❖ **To connect two laptops directly using Ethernet, traditionally you would use an Ethernet crossover cable.**

- **Crossover cable**: A crossover cable **swaps the transmit and receive wires**. this allows one laptop to send data while the other laptop receives it.

- **Straight-through cable: A straight-through cable is normally used to connect a laptop to a switch or router.**

4. **List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.**

Device	**Ethernet Standard**	**Use**
Wi-Fi Router	1000BASE-T	Connects to PC or other network devices
Desktop PC	1000BASE-T	Provides fast wired internet
Network Printer	100BASE-TX	Connects the printer to the college/home network